Laboratory Out of Specification Protocol

LABORATORY OOS INVESTIGATION PROTOCOL

Company : **Division**

OOS No. : **RefNo**

Date Initiated : **DateSent**

Issue by : NoIdRequestor1 / **Requestor1**

CC : CC

Name of analyst reported OOS : NameAnalyst

Details of OOS Laboratory Results

Product Name : ProductName

Batch No. : BatchNo

Date of Manufacturing : ManuDate

Description : Desc

Attachment : Attachment

PHASE I LABORATORY INVESTIGATION

PHASE 1A: PRELIMINARY INVESTIGATION

Please tick whichever is applicable

1	**Reference record/ document review**		**Remarks**
	1.1 Was the correct specification followed? Spec code No. : D_1	Q1_1	R_1
	1.2 Was the correct method of analysis followed? Method doc No. : D_2	Q1_2	R_2
	1.3 Was the method validated? Method validation doc No. : D_3	Q1_3	R_3
2	**Sampling procedure review**		
	2.1 Was the sampling procedure followed as per repective WI? Sampling procedure : D2_1	Q2_1	R2_1

	2.2 Use of proper and clean sampling tools & container?	Q2_2	R2_2
	2.3 Was the sample labelled correctly?	Q2_3	R2_3
3	**Sample condition checking**		
	3.1 Was the sample stored as per required storage condition?	Q3_1	R3_1
	3.2 Was the physical appearance of the sample typical?	Q3_2	R3_2
	3.3 Was the correct sample used for analysis?	Q3_3	R3_3
	3.4 Was there any sample mix up during storage/ handling?	Q3_4	R3_4
4	**Sample preparation review**		
	4.1 When were the standard and sample prepared?	Q4_1	R4_1
	4.2 Were the glasswares and lab apparatus used clean from contamination?	Q4_2	R4_2
	4.3 Was the correct pipette/ volumetric flask/ other glasswares used for dilution as per mentioned in the test procedure?	Q4_3	R4_3
	4.4 Was the proper chemical/ reagent/ volumetric solution used?	Q4_4	R4_4
	4.5 Was the physical condition of the chemical/ reagent/ standard solution and its validity found satisfactory? Description : D4 Expire date : D4_1	Q4_5	R4_5
	4.6 Was the standardization of the volumetric solution performed adequately?	Q4_6	R4_6
	4.7 Were the calculations for the volumetric solution verified for correctness?	Q4_7	R4_7
	4.8 Was there any mishandling of sample such as spilling of sample solution or incomplete transfer of a sample composite?	Q4_8	R4_8
	4.9 Was any abnormality observed during sample preparation?	Q4_9	R4_9
	4.10 Was it checked whether the analyst performing multiple testing which cause potential	Q4_10	R4_10

	sample mix up or cross contamination?		
5	**Working/ reference standard review**		
	5.1 Was the correct working/ reference standard used? Working/ reference standard description : D5_1_1 Supplier : D5_1_2	Q5_7	R5_7
	5.2 When was the working/ reference standard prepared or opened? Date opened/ prepared : D5_1_3	Q5_8	R5_8
	5.3 Was the working/ reference standard still within its expiry date? Batch no : D5_1_4 Expire date : D5_1_5	Q5_9	R5_9
	5.4 Was the working/ reference standard stored in proper condition?	Q5_10	R5_10
	5.5 Was the physical appearance of the working/ reference standard found satisfactory?	Q5_11	R5_11
6	**Instruments review**		
	6.1 Was the instrument calibrated? (I) Name of instrument/ ID : D5_1 Date of calibration : D5_2 Expire date : D5_3	Q5_1	R5_1
	(II) Name of instrument/ ID : D5_4 Date of calibration : D5_5 Expire date : D5_6		
	6.2 Was the correct method setting being used to analyse the sample?	Q5_2	R5_2
	6.3 Was the instrument equilibrated / cleaned properly before used?	Q5_3	R5_3

	6.4 Was any abnormality observed on the instrument during sample analysis?	Q5_4	R5_4
	6.5 Was there any instrument error or abnormality reported for the last 1 month?	Q5_5	R5_5
7	**Raw data review**		
	7.1 Were there any deviations from the analytical test procedure (eg: sample weight, dilution)?	Q6_1	R6_1
	7.2 Were the calculation and formula used correctly?	Q6_2	R6_2
	7.3 Is the GC chromatogram integrated properly?	Q6_3	R6_3
	7.4 Was any evidence of contamination from the previous injection?	Q6_4	R6_4
8	**Analyst review**		
	8.1 Was the analyst trained on the test method and performance record found satisfactory?	Q7_1	R7_1
9	**Other review**		
	9.1 Any other batches tested at that period of time?	Q8_1	R8_1
	9.2 Were these batches results found satisfactory?	Q8_2	R8_2

SUMMARY OF PHASE 1A PRELIMINARY INVESTIGATION (tick whichever applicable)

SPI1A

PHASE 1B: HYPHOTHESIS STUDY/ EXTENDED LAB INVESTIGATION

Note :-
a) Hypothesis results shall not be considered for reporting.
b) Re-test for section 10.1 to 10.3 shall be performed using ori:ginal sample.
c) Re-sampling from the same lot of the initial test shall be done in the following cases but not limited to:-
(i) If the original sample exhausted
(ii) To perform any hyphotesis study

10	**10.1 Sample preparation error/ test solution is suspected?**	Q10	R_10
	10.1.1 Contaminated/ decomposed volumetric solution/ reagent is suspected?	Q10_1	R_10_1
	10.1.1 Restandardization of volumetric solution is needed?	Q10_2	R_10_2
	10.1.2 New sample preparation is needed?	Q10_3	R_10_3

10.1.3 Need to retest?	Q10_4	R_10_4
10.1.4 Test result		
Test	**Solution 1 (Initial)**	**Solution 2**
#1	Q10_5_1	Q10_5_6
#2	Q10_5_2	Q10_5_7
#3	Q10_5_3	Q10_5_8
Mean	Q10_5_4	Q10_5_9
Attachment	Q10_5_5	
10.2 Instrument malfunction is suspected?	Q10_6	R_10_5
10.2.1 Need to recalibrate the instrument?	Q10_7	R_10_6
10.2.2 Need to retest the sample by other instrument?	Q10_8	R_10_7
10.2.4 Test result		
Test	**Instrument 1**	**Instrument 2**
#1	Q10_6_1	Q10_6_6
#2	Q10_6_2	Q10_6_7
#3	Q10_6_3	Q10_6_8
Mean	Q10_6_4	Q10_6_9
Attachment	Q10_6_5	
10.3 Analyst error is suspected?	Q10_10	R_10_8
10.3.1 Need to assign analyst 2 to conduct the test?	Q10_11	R_10_9
10.3.2 Test result		
Test	**Analyst 1**	**Analyst 2**
#1	Q10_7_1	Q10_7_6
#2	Q10_7_2	Q10_7_7
#3	Q10_7_3	Q10_7_8
Mean	Q10_7_4	Q10_7_9
Attachment	Q10_7_5	
10.4 Improper sampling/ sample contamination is suspected?	Q10_12_1	Q10_6_6_1
10.4.1 Sampling tools not clean?	Q10_12_2	Q10_6_6_2
10.4.2 Suspected deviation from sampling procedure i.e.not enough rinsing	Q10_12_3	Q10_6_6_3
10.4.3 Need to retest to confirm	Q10_12_4	Q10_6_6_4

whether sample is contaminated/ sampling procedure deviation?		
10.4.4 Test result		
Test	**Original Sample**	**New Sample**
#1	Q10_8_1	Q10_8_6
#2	Q10_8_2	Q10_8_7
#3	Q10_8_3	Q10_8_8
Mean	Q10_8_4	Q10_8_9
Attachment	Q10_8_5	

SUMMARY OF PHASE 1B HYPOTHESIS STUDY (tick whichever applicable)

SPI1B

CONCLUSION FOR PHASE I LABORATORY INVESTIGATION

A: Assignable cause / root cause:

rootcause

B: Correction action taken (if assignable cause is identified):

i) If the OOS results is due to reporting error i.e. calculation error/ laboratory error/ wrong peak integration, rectify the error based on the finding.
Correction action taken:

correctionaction

Note:

Note1

ii) If the error due to analytical error, analyst I to correct the cause and perform retesting of the sample in triplicate with new sample preparation.
Correction action taken:

correctionaction_1

Test	**Result**
#1	CR_1
#2	CR_2
#3	CR_3
Mean	CR_4

Tested by: testby

Date : testdate

Note:

Note2

C: Final conclusion of the investigation

conclusion

Batch is **Batch**

Please give your justification if the batch is rejected.

B_1

OOS is due to laboratory/measurement error **Cause**

(If Yes, please raise OOSR OOSR*, or the investigation is ended)*

PHASE II INVESTIGATION

PHASE 2A - PRODUCTION REVIEW/ CROSS-FUNCTIONAL INVESTIGATION

1	**Raw Material/ Feedstock**		**Remarks**
	1.1 Was any raw material used in manufacturing released under deviation?	Q2	R2
2	**Plant Process**		
	2.1 Were the in-process checks carried out as per PQP?	Q2_4	R2_4
	2.2 Were the in-process checks results within specification?	Q2_5	R2_5
	2.3 Was there any breakdown during production process?	Q2_6	R2_6
	2.4 Were utilities(e.g: steam water, temperature, pressure, etc. are as per the requirement throughout the production process)?	Q2_7	R2_7
	2.5 Were the product stored properly in the storage tank?	Q2_8	R2_8
	2.6 Were the storage tank / SPDLI results within specification?	Q2_9	R2_9
	2.7 Was the product being contaminated during transferring?	Q2_10	R2_10
	2.8 Was the cleaning carried out during product changeover? Cleaning method : R2_14 Previous production : R2_15	Q2_11	R2_11
	2.9 Was the correct sampling procedure is used?	Q2_12	R2_12
3	**History**		
	3.1 Have there been similar errors in the past?	Q2_13	R2_13

SUMMARY FOR PRODUCTION REVIEW/ CROSS-FUNTIONAL REVIEW (tick whichever applicable)

SPR

PHASE 2B - ADDITIONAL LABORATORY TESTING

(i) RETESTING

Note:-
1. Section head/ designee shall allot retesting of the original sample to analyst 2 in triplicate if required.
2. The sample used for the retesting should be taken from the same homogeneous material that was collected from the lot tested and yielded the OOS results.
3. If all the results tested by analyst 2 are complying,designee shall allot retesting of the same sample by analyst 1 in triplicate.
4. Maximum 3 retest in triplicate by 3 different analyst(including analyst 1) is allowed.

a) Retesting in triplicate by analyst 2 on the original sample.

Test	Result
#1	CR_1_1
#2	CR_2_1
#3	CR_3_1
Mean	CR_4_1

Tested by: testby_1

Date : testdate_1

b) Retesting in triplicate by analyst 1 on the original sample.

Test	Result
#1	CR_1_2
#2	CR_2_2
#3	CR_3_2
Mean	CR_4_2

Tested by: testby_2

Date : testdate_2

Note:
a) If all the six determinations value (individual results and mean) are within the specification, report the mean value and consider the OOS test result as invalid.

b) Any results (individual results and mean) are out of specification, then consider the OOS test result is valid. The batches should be rejected / can be held for further investigation. Online OOSR shall be initiated.

Conclusion:

Mean of the 6 determinations : mean6

(ii) RESAMPLING

Note:-
1. Resampling is only allowed if it is proven that the original sample was non-representative /or if found that sampling procedure adopted was wrong.
2. Resampling involves analysing of sample from any additional units collected as part of the original sampling procedure or from a new sample collected from the batch.
3. Designee shall allot the retesting on the new resampled sample to analyst 1 and analyst 2 in triplicate.

Justification of sampling : Justification

a) Resampling and retesting in triplicate by analyst 2.

Test	Result
#1	CR_1_1_1
#2	CR_2_1_1
#3	CR_3_1_1
Mean	CR_4_1_1

Tested by: testby_1_1

Date : testdate_1_1

b) Resampling and retesting in triplicate by analyst 1.

Test	Result
#1	CR_1_2_1
#2	CR_2_2_1
#3	CR_3_2_1
Mean	CR_4_2_1

Tested by: testby_2_1

Date : testdate_2_1

Conclusion : ConclusionResamp

Mean of 6 determinations : mean6deter

Note :-
a) If all the six determinations value (individual results and mean) are within the specification, report the mean value and consider the OOS test result as invalid.
b) Any results (individual results and mean) are out of specification, then consider the OOS test result is valid. The batches should be rejected/ can held for further investigation. Online OOSR shall be initiated.

SUMMARY FOR ADDITIONAL LABORATORY TESTING (tick whichever applicable)

SALT

CONCLUSION FOR PHASE II INVESTIGATION

A: Assignable cause/ root cause:

rootcause2

B: Correction action taken (if assignable cause is identified):

correctiveaction2

C: Final conclusion of the investigation

conclusion2

Batch is **status**

Please give your justification if the batch is rejected.

justification2

OOS is due to laboratory/measurement error **Cause_1**

(If Yes, please raise OOSR OOSR*, or the investigation is ended)*

INVESTIGATION DONE BY :-

Name : InvName

Designation : InvDesg

Date : InvDate

APPROVED BY :-

APPROVE REJECT

Name : AppName

Designation : AppDesg

Date : AppDate

Remarks : AppRem

App

New OOSR

QCD-144/010323/02

Hidden Fields:-

Current Action	: CA
Requestor	: DocAuthor
Date Created	: CDate
Verificator (Chemists)	: QCVerificator
To lock the Imp	

	: LockSectionImp
To lock the Eff	: LockSection2
QC HOD	: QCHOD
Editor1	: AEditor1
Editor2	: AEditor2
Editor3	: *AEditor3*
Editor4	: AEditor4
Editor5	: AEditor5
increment	: increment
Mailing List - All Recipients	: All
CAStatus - Verification	: CAStatus
IT ADMIN	: ISADMIN
INO	: Year - Month - CNO